

What Is Pressure?

Measuring the invisible push: force per unit area

Today we get our first tool for explaining the crushed jerrycan.

Something pushed hard enough to crush it.

- Look at the crushed jerrycan from last lesson. List every force you think is acting on it right now, and show its direction with an arrow.
- Why do you think the outside of the jerrycan “won” against the plastic?
- Write silently for 1 minute, then compare with your partner.

Draw your arrows now. We will check them at the end of the lesson.

Same force. Same mass. Will they feel the same?

A flat textbook

Pressed onto your palm with a firm push.

A sharpened pencil

Same mass as the book, pressed onto your palm with exactly the same push.

Write a 2-sentence prediction before your teacher demonstrates. Which will feel like more of a “push”, and why?

Pressure is force spread over an area.

- Pressure = Force $\div$ Area $P = F / A$
- Force (F) is measured in newtons (N). Area (A) is measured in square metres (m²).
- Pressure is measured in pascals (Pa). 1 Pa = 1 N/m²
- The pencil hurts more because the same force is squeezed onto a tiny area, so the pressure is much bigger.

QUICK CHECK 1

A book and a pencil are pressed onto your palm with the same force. Why does the pencil tip hurt more?

- A** The pencil is heavier than the book
- B** The same force acts on a much smaller area, so the pressure is greater
- C** The pencil pushes with a bigger force
- D** The book spreads the force over a smaller area

Answer key is teacher-side only, in a separate file.

How much pressure do you put on the floor?

1 Stand on the bathroom scale. Record your mass in kg. Convert to weight in newtons: $W = m \times 10$ ($g = 10 \text{ N/kg}$).

2 Stand on graph paper with ONE shoe. Your partner traces around the sole.

3 Count the 1 cm^2 squares inside the outline. Count a part-square only if more than half of it is inside.

4 Force on one foot = half your weight. Convert area: $\text{cm}^2 \div 10\,000 = \text{m}^2$. Then $P = F / A$.

Record your shoe type, area, force and pressure. Post them on the class data wall.

A 60 kg student in school shoes

1. Weight = $60 \text{ kg} \times 10 \text{ N/kg} = 600 \text{ N}$. One foot carries half: $F = 300 \text{ N}$.
2. Sole outline = 200 squares = 200 cm^2 . $A = 200 \div 10\,000 = 0.02 \text{ m}^2$.
3. $P = F / A = 300 \text{ N} \div 0.02 \text{ m}^2 = 15\,000 \text{ Pa}$.

Most common mistake: forgetting to change cm^2 into m^2 before dividing. Check yours!

QUICK CHECK 2

A shoe sole covers 150 cm^2 . What is this area in square metres?

A 1.5 m^2

B 0.15 m^2

C 0.015 m^2

D 15 m^2

Answer key is teacher-side only, in a separate file.

Which shoes press hardest on the floor?

Large contact area

Flat sandals and sports trainers spread your weight over a big area → lower pressure.

Small contact area

Heels and narrow soles put your weight on a small area → higher pressure. That is why a heel can dent a soft floor.

Same person, different shoes: the force stays the same, only the area changes.

A 90 kg maize bag sits on a pallet with 4 legs.

1. Weight of bag = $90 \text{ kg} \times 10 \text{ N/kg} = 900 \text{ N}$, shared by 4 legs $\rightarrow 225 \text{ N}$ per leg.
2. Each leg base = $25 \text{ cm}^2 = 25 \div 10\,000 = 0.0025 \text{ m}^2$.
3. Pressure under each leg = $225 \text{ N} \div 0.0025 \text{ m}^2 = 90\,000 \text{ Pa}$.

QUICK CHECK 3

The pressure under each pallet leg is 90 000 _____. Which unit completes the answer?

A Newtons (N)

B Pascals (Pa), which are N/m^2

C Kilograms (kg)

D Square metres (m^2)

Answer key is teacher-side only, in a separate file.

Same 150 N push. Which is safer on a soft floor?

A rungu

A Maasai elder leans on a rungu. The handle tip touches the floor over just 2 cm^2 .

A crutch

A patient at Kenyatta National Hospital leans on a crutch with a flat base of 50 cm^2 .

Calculate both pressures on your mini-whiteboard. Then explain which is safer, and why.

QUICK CHECK 4

A rungu tip (2 cm^2) and a crutch base (50 cm^2) each press down with 150 N. Which statement is correct?

- A** They exert the same pressure because the force is the same
- B** The crutch exerts higher pressure because it is larger
- C** The rungu exerts higher pressure because its area is smaller
- D** Neither exerts pressure because they are not moving

Answer key is teacher-side only, in a separate file.

Air pushes on every surface, from every direction.

- The air above us has weight. That weight presses on every surface it touches: this is atmospheric pressure.
- It pushes on ALL faces of the jerrycan — top, bottom and all four sides — always pointing inward.
- Air inside the jerrycan pushes outward. While the two pushes balance, the jerrycan keeps its shape.
- Next lesson we find out how big atmospheric pressure really is, and what happened when the balance broke.

QUICK CHECK 5

In which directions does atmospheric pressure push on the jerrycan?

- A** Only downward, on the top
- B** Only on the sides
- C** Inward on every surface: top, bottom and all sides
- D** Outward from the inside only

Answer key is teacher-side only, in a separate file.

Update your model. Add one new question.

UPDATE YOUR MODEL

Redraw your jerrycan. Add arrows on ALL six faces pointing inward (air outside), and arrows pointing outward from inside. Finish the sentence: "The full jerrycan did not collapse because ____." Circle what changed since Lesson 1 in green.

DRIVING QUESTION BOARD

Yellow note: What do I now THINK is crushing the jerrycan? Use the words pressure, force and area.

Blue note: What am I still WONDERING? For example: how strong is atmospheric pressure? Does water push the same way?

From the ARES offline library



VIDEO [Pressure and Pascal's principle \(part 2\)](#)

Khan Academy • Good match for force, area and pressure



READING [Pressure versus Temperature \(PLIX\)](#)

CK-12 • Weak match: about gas pressure and temperature, not $P = F/A$. Review before using.



SEARCH [Search ARES: pressure, force, unit area](#)

Opens an ARES search for related videos and notes

Links open only on the school network (ares.local). Click to open during the lesson.

5 questions asked during class today

1. Why does the pencil tip hurt more than the book?
2. Convert 150 cm^2 to m^2 .
3. The pressure under each pallet leg is 90 000 _____. Which unit?
4. Rungu (2 cm^2) vs crutch (50 cm^2), both 150 N: which is correct?
5. In which directions does atmospheric pressure push on the jerrycan?

Ask your teacher how to submit your answers today (for example, on a paper answer slip).